

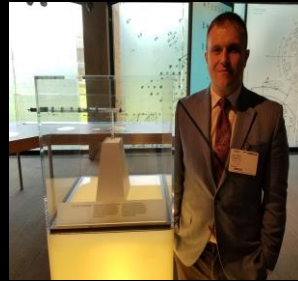
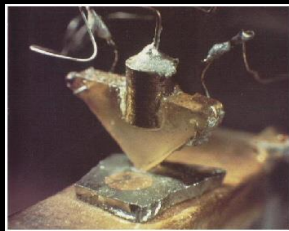
Cadence Semiconductor 101: Basics of Semiconductors

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University of Notre Dame
December 2, 2024



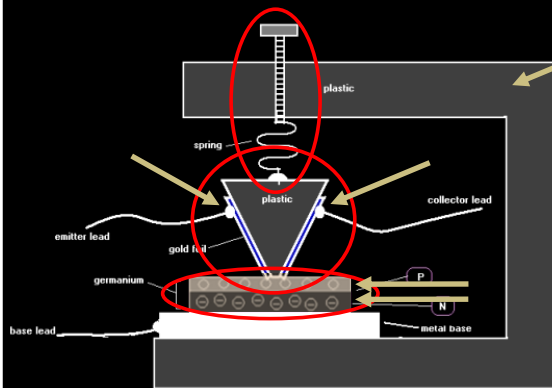
1947: The Point Contact Transistor

- John Bardeen and Walter Brattain at Bell Labs
 - Amplify electric signals
 - *Composed entirely of cold, solid substances*



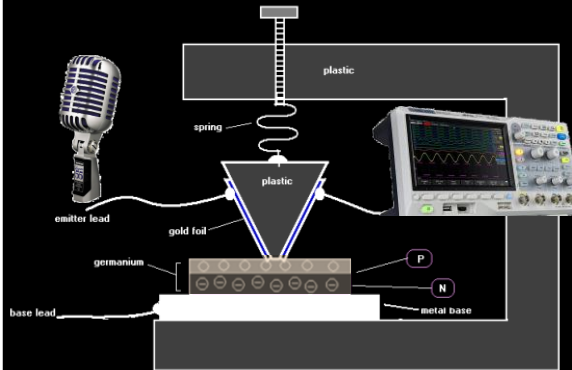
From Bardeen and Brattain: *It is, if you will, the electrical equivalent of a vacuum tube amplifier. But there the similarity ceases. It has no vacuum, no filament, no glass tube. It is composed entirely of cold, solid substances."*

How did these Transistors work?



- A plastic stand to hold everything
- A chunk of germanium
 - N-type – Will explain in a moment
 - P-type – Will explain in a moment
- Plastic triangle
 - Gold foil along the sides
 - Spring to push point of triangle into the germanium

How did these Transistors work?

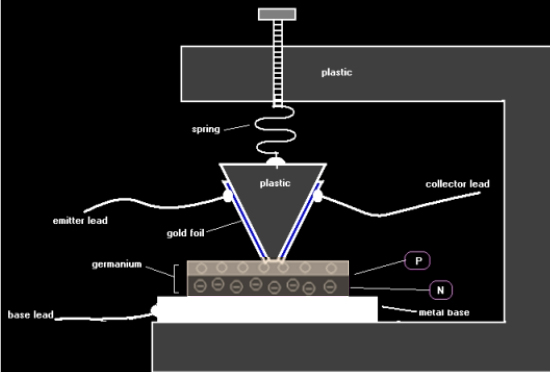


- December 16, 1947
 - Microphone on the emitter
 - Oscilloscope to the collector
 - Voice signal got *amplified*
 - “The voice went over the circuit... A distinct gain in speech level could be heard and seen on the scope.”

ACADEMIC NETWORK



How did these Transistors work?



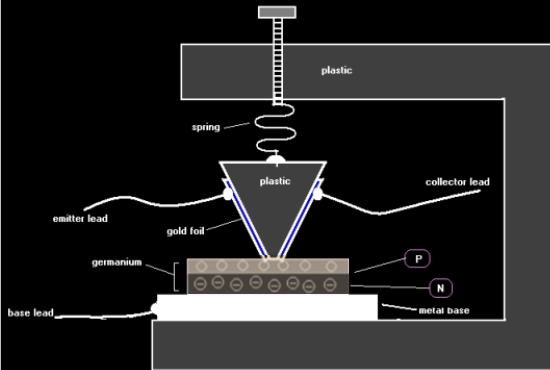
- Wait, how?
 - The key is in the germanium
- Three ways to classify materials in how they conduct electricity
 - **Conductor** - Metals that readily transport electricity
 - **Insulators** – Refuse to allow current to flow

🍀 **Third type: Semiconductors**

ACADEMIC NETWORK



How did these Transistors work?



- Semiconductors
 - Conduct better than insulators
 - Not as well as conductors
- Two Ways to Conduct
 - Negative Charge Carriers
 - Positive Charge Carriers

ACADEMIC NETWORK



Course Acronyms

V - Very

L - Large

S - Scale

I - Integration

C - Complementary

M - Metal

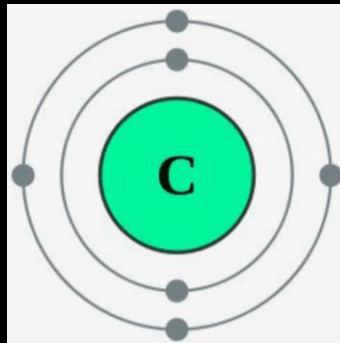
O - Oxide

S - Semiconductor



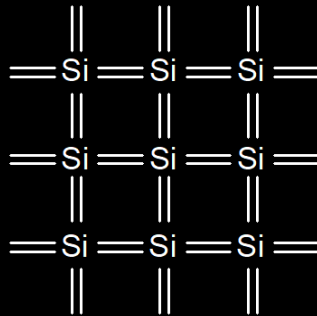
Chemistry Review: Group IV elements

- Has **four** electrons and **four** holes
 - Achieves a full outer energy level by forming **four covalent bonds**



Lattice

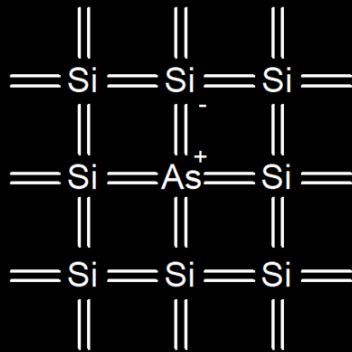
- Group IV atoms
 - Forms crystal lattice with bonds to **four neighbors**



N-Type Semiconductor

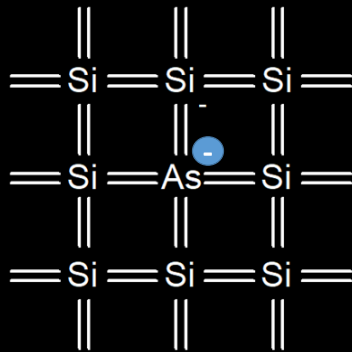
- Dope a lattice with Group V element (As): More **N**egative

❖ 5th valence electron is loosely bound



N-Type Semiconductor: Borrowing

- **Thermal vibrations** - sets the electron free
 - Leaves a **positively charged As⁺ ion** and a **free electron**



Electrons *carry*
the current

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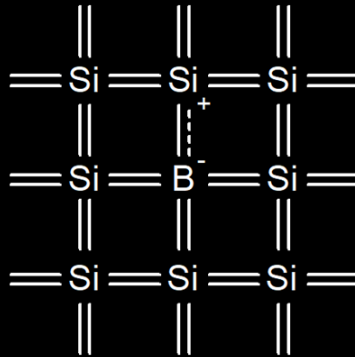
O - Oxide

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P-Type Semiconductor

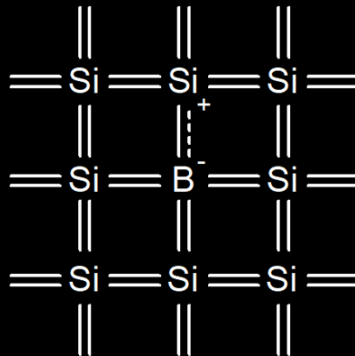
- Dope a lattice with Group III element (B): More **P**ositive



P-Type Semiconductor: Borrowing

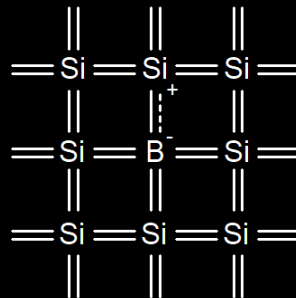
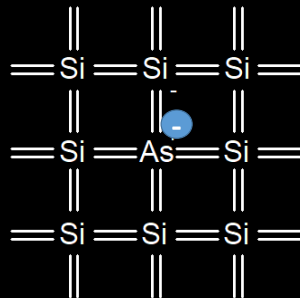
- Leaves a negatively charged B⁻ ion and a free hole in Si
 - Silicon atom can now borrow another e⁻ from another atom

-



N-Type and P-Type Semiconductors

- Pure Group IV has **no free carriers** and **conducts poorly**
 - Adding dopants **increases the conductivity**



Course Acronyms

V - Very

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C - Complementary

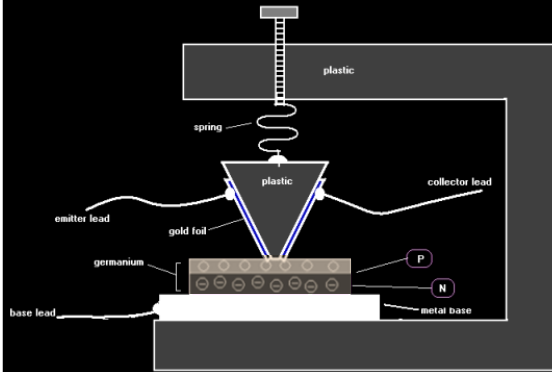
M - Metal

O - Oxide

S - Semiconductor

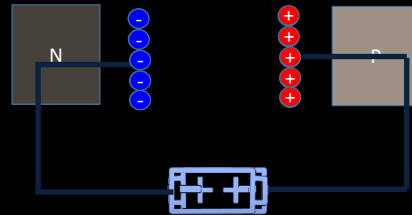


PN Junctions



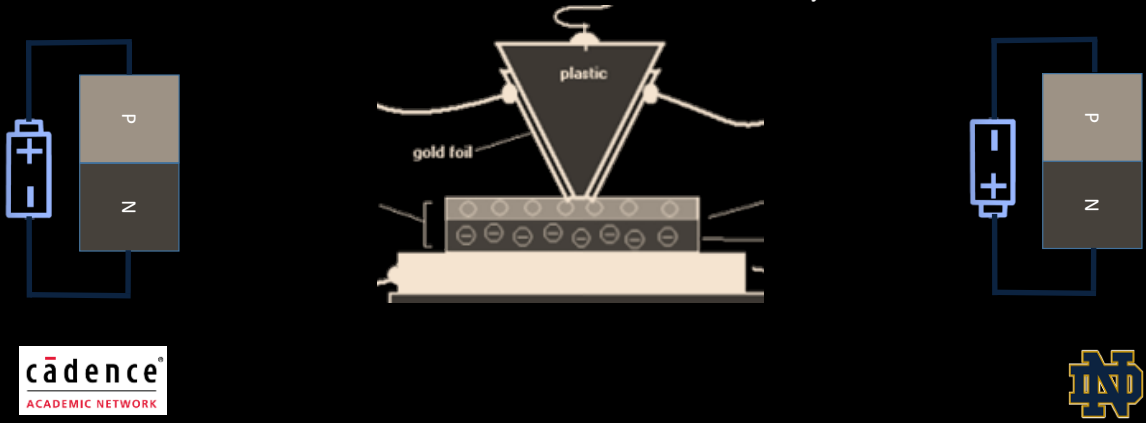
ACADEMIC NETWORK

- + and – Charge Carriers
 - Device that flows in *only* one direction: Electrons to Holes



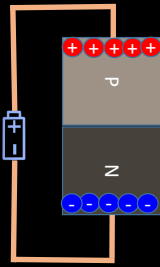
How did these Transistors work?

- It's best to think of this device as two one-way current valves

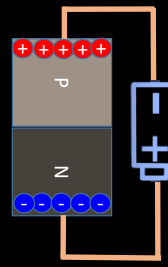


How did these Transistors work?

Small Battery
on left allows
current to flow

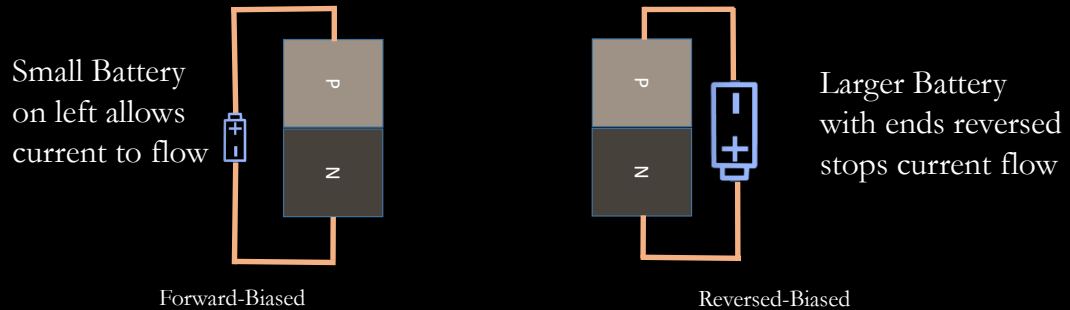


Larger Battery
with ends reversed
stops current flow



How did these Transistors work?

- When they were put together, the world changed...



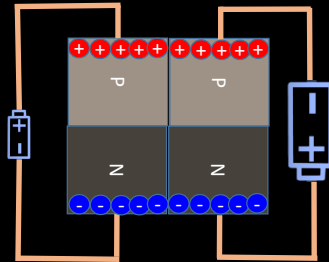
The key was in the gold film

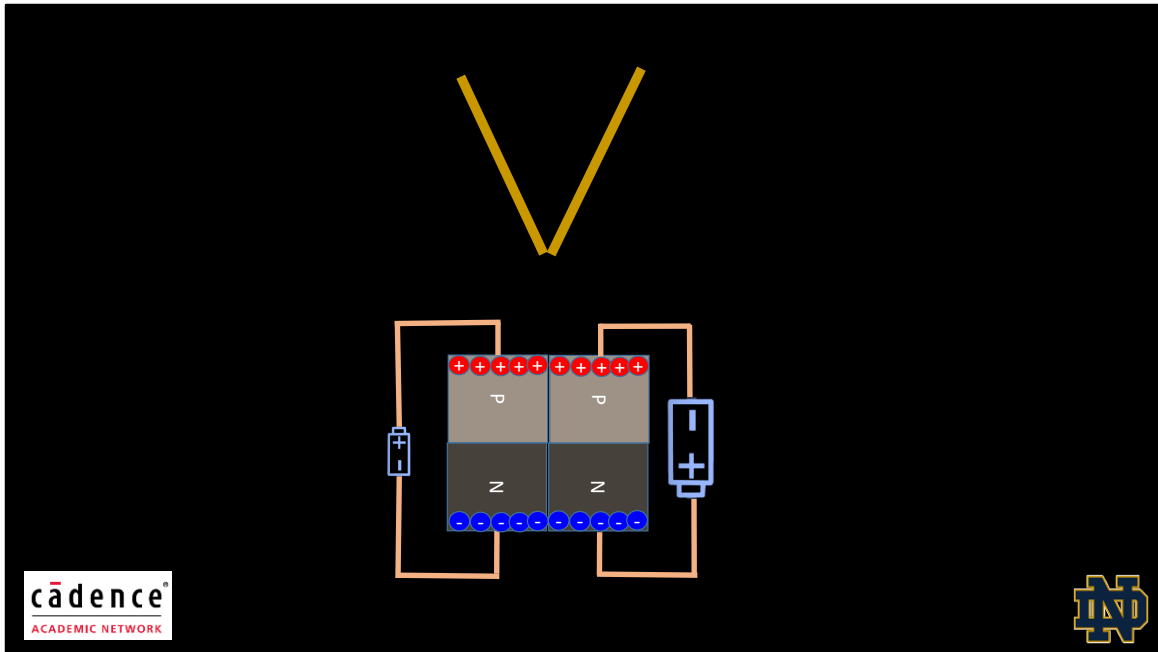
- Made a connection between two “halves”
 - Injected holes into the opposite side



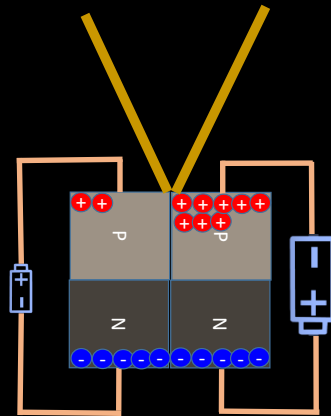
How did these Transistors work?

- Distance between the Gold Contacts is $2/1000^{\text{th}}$ of an inch
 - Positive carriers “injected” into the reverse-biased diode





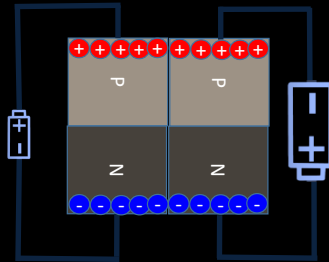
Now the holes are attracted



- Now there is a difference in *potential*
- There is now opportunity for current to flow on the right

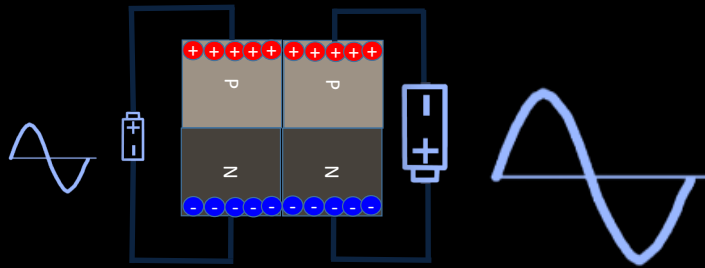
How did these Transistors work?

- The current on the *right* is now controlled by the *left*
 - Recall that Power is current times *voltage*



How did these Transistors work?

- The current on the *right* is now controlled by the *left*
 - $i_{\text{forward}} \approx i_{\text{reverse}} \dots$ but $V_{\text{reverse}} > V_{\text{forward}}$



How did these Transistors work?

- Actual measurements from the very first transistor experiment

$$I_{\text{forward}} = 0.75 \text{ mA}$$

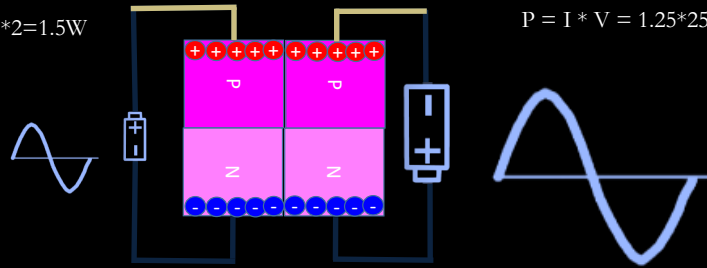
$$V_{\text{forward}} = 0.2 \text{ V}$$

$$P = I * V = 0.75 * 2 = 1.5 \text{ W}$$

$$I_{\text{reverse}} = 1.25 \text{ mA}$$

$$V_{\text{reverse}} = 25 \text{ V}$$

$$P = I * V = 1.25 * 25 = 31.25 \text{ W}$$



Course Acronyms

V - Very

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M - Metal

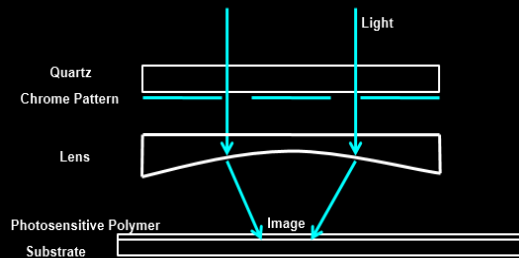
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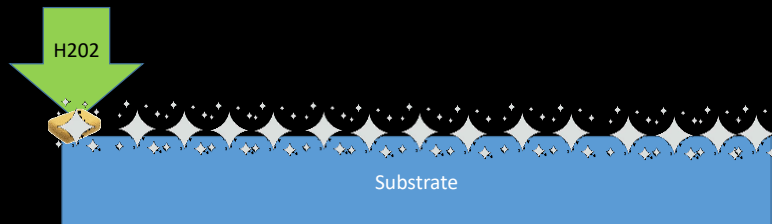
Lithographic Process

- CMOS Photolithography
 - Uses ultra-violet light to etch into photosensitive polymers



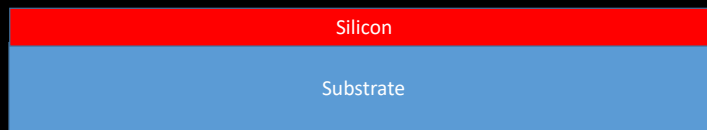
Lithographic Process

- First step – Substrate Cleaning
 - H_2O_2 is used to clean the substrate of any contaminants that may have been on the wafer



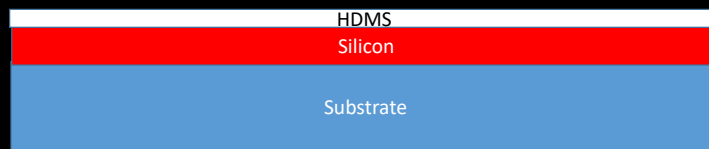
Lithographic Process

- Second Step – Silicon Layer
 - Add a layer of silicon on top of the substrate



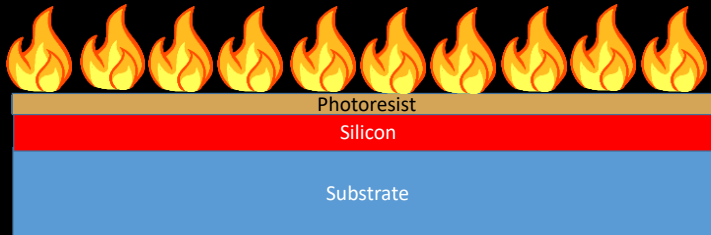
Lithographic Process

- Third Step – Add Adhesive
 - HDMS (Hexamethyldisilazane) applied to increase adhesion of photoresist on the silicon wafer surface



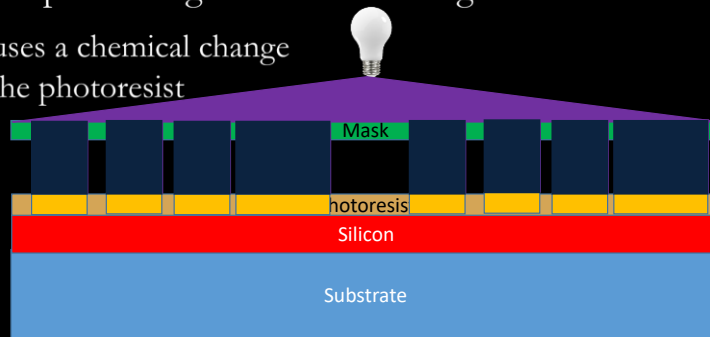
Lithographic Process

- Fourth Step – Photoresist Coating / Prebake
 - An organic photoresist that softens where exposed to light is spun and baked onto the wafer



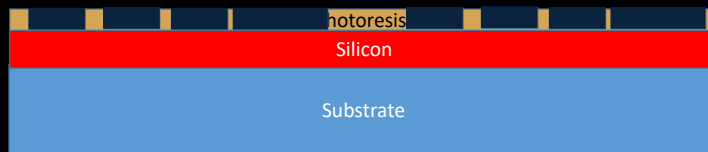
Lithographic Process

- Fifth Step –UV Light is shone through a mask
 - Causes a chemical change in the photoresist



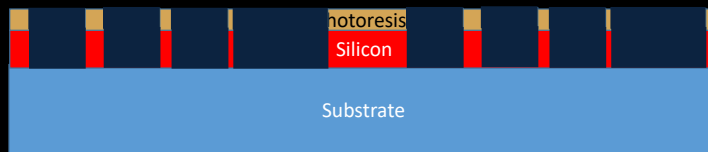
Lithographic Process

- Sixth Step –Development
 - Exposed photoresist is now soluble, and can be washed away
 - ❖ Called *piranha etch*



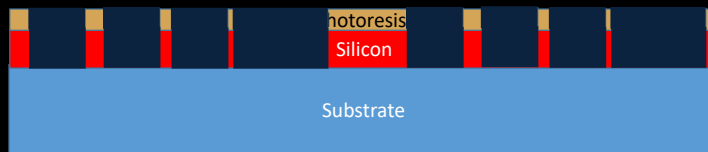
Lithographic Process

- Seventh Step – Etching
 - Use a chemical agent to remove the unprotected silicon



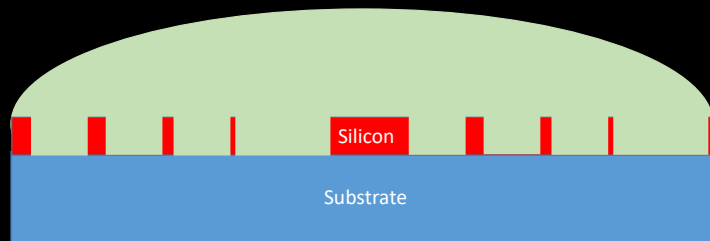
Lithographic Process

- Eighth Step – Remove insoluble photoresist
 - Use a chemical edge stripper to remove the remaining photoresist



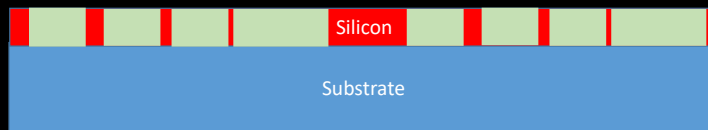
Lithographic Process

- Ninth Step – Add Silicon Dioxide SiO_2



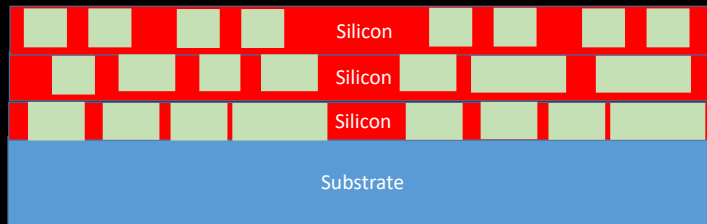
Lithographic Process

- Tenth Step – Polish to remove excess silicon dioxide



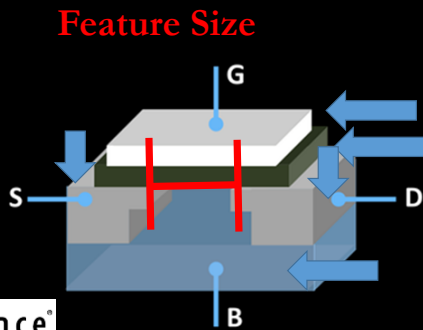
Lithographic Process

- Next several steps – Repeat as many layers as necessary to get the desired design



Innovation of Transistors

- MOS – Field Effect Transistor (MOSFET)



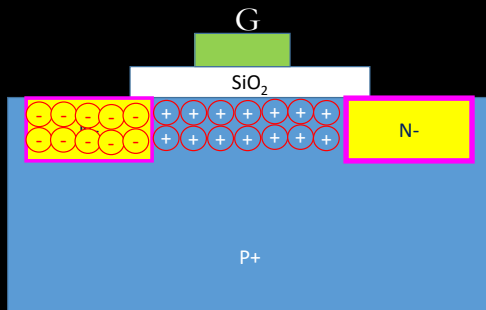
cadence
ACADEMIC NETWORK



- Gate – A polysilicon conductor layer
 - ❖ insulating layer of SiO₂.
- Source – The input pin.
- Drain – The output pin.
- Body – The Silicon wafer itself

Innovation of Transistors

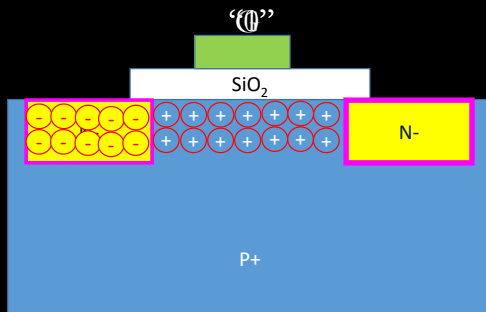
- nMOS (n-Type MOSFET)



- Source and drain have the opposite doping as the body
- ❖ SiO₂ serves as a thin layer between the body and the gate metal

Innovation of Transistors

- nMOS (n-Type MOSFET)



- Voltage is put on the Gate

- ❖ Logic “0”

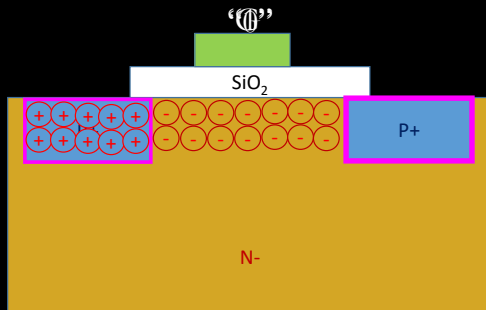
- Holes attracted to -, stopped by SiO2. No current

- ❖ Logic “1”

- Holes repelled by +. Electrons attracted. Stopped by SiO2 => Go to Drain => Current Flow

Innovation of Transistors

- pMOS (p-Type MOSFET)



- Voltage is put on the Gate

- ❖ Logic “1”

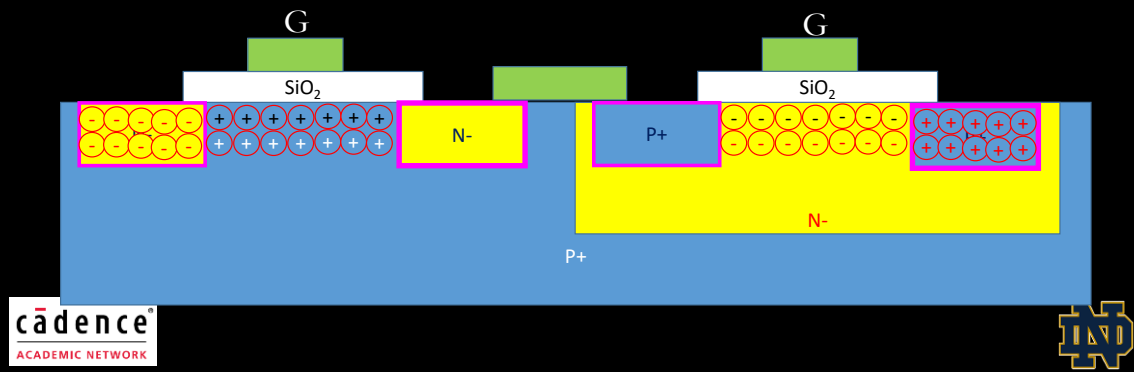
- Holes attracted to +, stopped by SiO2. No current

- ❖ Logic “0”

- Holes repelled by -. Holes attracted. Stopped by SiO2 => Go to Drain => Current Flow

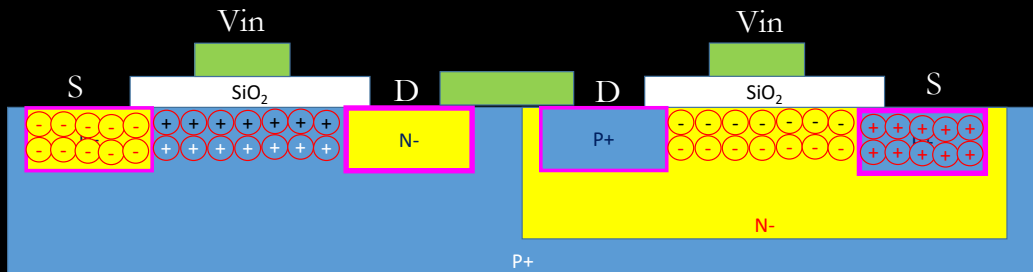
What if we put them together?

- Gate (G), Source (S), Drain (D)



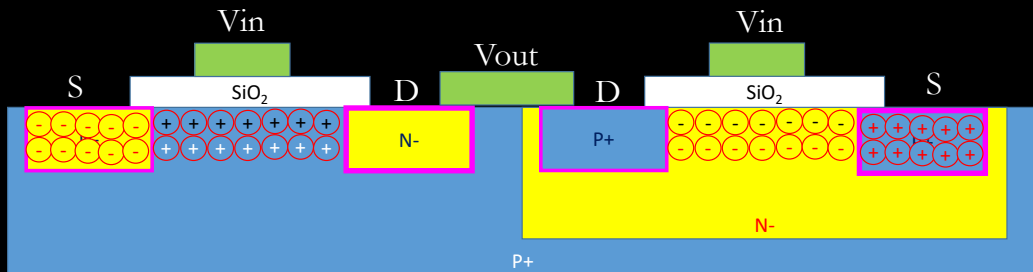
What if we put them together?

- Tie an **input voltage** to both **gates**



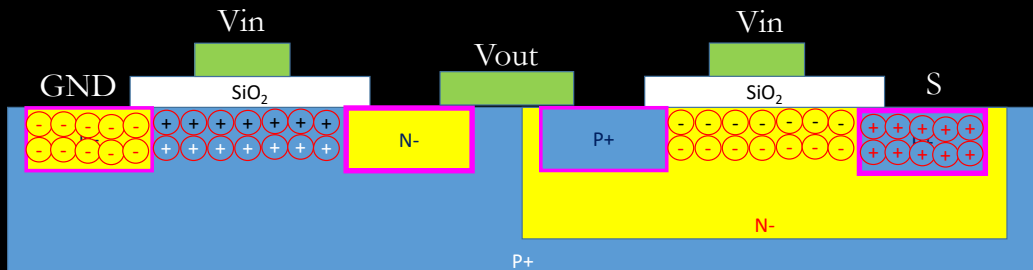
What if we put them together?

- Tie the **output voltage** to both **drains**



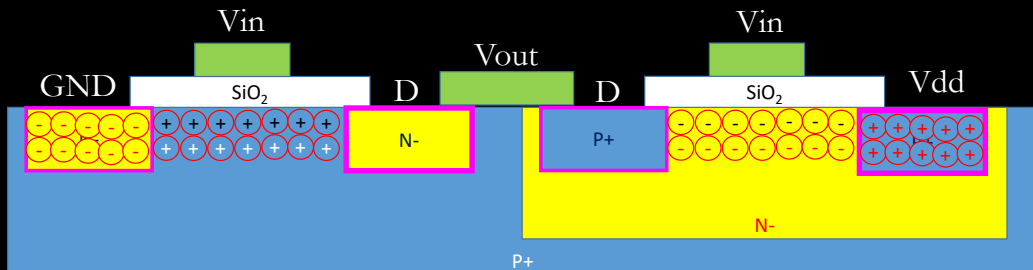
What if we put them together?

- For the **nMOS**, tie **Source** to **Ground** (“Logic 0”)



What if we put them together?

- For the **pMOS**, tie **Source** to **Supply Voltage** (“Logic 1”)



What if we put them together?

- What will happen if we put a “1” on V_{in} ?

